DEPARTMENTOFAPPLIEDMATHEMATICSANDCOMPUTATIONALSCIENCE
B. Tech. II YEAR (4YDC) SEMESTER-III
INFORMATION TECHNOLOGY ENGINEERING
MA***: APPLIED MATHEMATICS**

HOURSPERWEEK			CREDITS			MAXIMUM MARKS				
T	P	TU	T	P	TU	THEORY		PRACTICAL		TOTAL MARKS
						CW	END SEM	SW	END SEM	
2	--	1	2	--	1	30	70	--	--	100

PRE–REQUISITES: Mathematics for Engineers and Mathematics for Data Science

COURSEOBJECTIVES

Enable the students to apply the knowledge of Mathematics in various engineering fields by making them

1. To develop the concept of partial differential equation with its application.
2. To introduce the concept of Fourier series, Fourier transform and Laplace Transform with their applications.
3. To solve the problems related to differential calculus and integral calculus using numerical methods
4. To present all usual basic concepts of graph theory, graph properties (with simplified proofs) and formulations of typical graph problems.
5. To study the basic of mathematical programming technique and their applications.

COURSEOUTCOMES

After completing this course student will be able to

CO1: Solve linear homogeneous partial differential equations of nth order and their applications.

CO2: Apply the fundamental concepts of Fourier and Laplace in integral transforms, signal decomposition, and differential equation solutions.

CO3: Apply the basic principles and importance of numerical methods in scientific and engineering computations

CO4: Apply graph theory-based tools in solving practical problems.

CO5: Evaluate the problems based on linear and non-linear programming.

COURSECONTENTS

THEORY

UNIT1 **Partial Differential Equations**: Formation of Partial Differential Equations, Partial Differential Equations of first order and first degree i.e., Pp+Qq=R, Linear Homogeneous Partial Differential Equation s of nth order with constant coefficient, Separation of Variables and its applications.

UNIT2 **Fourier Analysis**: Fourier series: Definition and Derivations, Odd and Even functions, Half-Range Series, Change of Scale, Overview of Fourier Integral and Fourier Transform.

Laplace Transform: Definition and properties, Laplace Transforms of derivatives, Inverse Laplace Transform: properties, Convolution Theorem and applications.

UNIT3 **Calculus of Finite Differences**: Difference table, Operators E and Δ, Newton's forward and backward interpolation formula for equal intervals, Lagrange's interpolation formula and divided difference method for unequal intervals, Numerical Differentiation and Integration (Trapezoidalrule, Simpson's1/3ruleand Simpson's3/8rule).

UNIT4 **Graph Theory:** Graphs–Definitions and basic properties. Isomorphism, Euler Circuits and Hamiltonian cycle. Digraphs. Trees-properties, spanning trees, Planer graphs. Shortest path problem, Dijkstra algorithm, Shortest spanning tree-Kruskal and prims algorithm.

UNIT5 **Optimization Techniques:** Linear Programming Problem: Simplex Method for Maximization and Minimization. Revised Simplex Method and Duality Theorem. Non-linear Optimization Technique: Kuhn-Tucker condition, Fibonacci Search Method.

ASSESSMENT

1. Internal Assessment for continuous evaluation, mid- term tests, tutorials ,class performance, etc.(30%).
2. End semester Theory Exam(70%).

TEXT BOOKS RECOMMENDED

1. Ramana BV, Higher Engineering Mathematics, Tata Mc Graw Hill Publishing Company Ltd., NewDelhi,2006.
2. E. G. Goodaire and Michael M. Permexter, Discrete Mathematics with Graph Theory
3. KantiSwarup,P.K.GuptaandManMohan,Operationsresearch,SultanChand&Sons, Educational publishers , NewDelhi

REFERENCE BOOKS

1. Balaguruswamy E., Numerical Methods, Tata McGraw-Hill Publishing Company Ltd.,NewDelhi .
2. H. K. Das, Higher Engineering Mathematics, S.Chand New Delhi.
3. Erwin. Kreyszig, Advanced Engineering Mathematics, 8thedition, John Willyand sons Publications, 1999.